

CUSTOMER FUTURE CLV PREDICTION

Complete End-to-End Machine Learning Project Documentation

Final Selected Model: XGBoost Baseline

Recorded Best Overall Results: RMSE = 2282.51 | R^2 = 0.6596 | MAE = 663.78

Project Overview

This project predicts Future Customer Lifetime Value (Future CLV) from historical transactional behaviour. The work follows a complete machine learning pipeline from customer-level data preparation and feature engineering to exploratory analysis, model comparison, prediction, and customer segmentation.

1. Problem Statement

- Businesses need to identify customers likely to generate high future value. The problem was formulated as a supervised machine learning regression task.

- **Target:** Future_CLV.

Learning type: Supervised regression.

2. Project Objectives

- Create customer-level features from transactional data.
- Engineer RFM and behavioural features.
- Create a Future CLV target from future-period transactions.
- Perform EDA, correlation analysis, and outlier analysis.
- Build mean and median baseline models.
- Train Random Forest and XGBoost models.
- Evaluate models using MAE, RMSE, and R².
- Analyse feature importance, residuals, and actual-vs-predicted behaviour.
- Predict customer Future CLV and segment customers into value groups.

3. Target Construction

- The transaction value was calculated as **Amount = Quantity × Price**.
- The future-period data was grouped by Customer ID and Amount was summed. The aggregated target was renamed **Future_CLV**.
- The target dataset contained **2,868 customers**.

4. Target Statistics

- Future_CLV was highly right-skewed. Most customers had relatively low future value, while a small number had extremely high values. This long-tail behaviour made the prediction task more difficult, especially for extreme customers.

5. Customer Feature Engineering

- The final customer-level feature table contained **3,313 customer records**.
- Features: Customer ID, Monetary, Frequency, Recency, Average_Order_Value, Total_Quantity, Unique_Products, Active_Months, and Avg_Purchase_Interval.

6. Exploratory Data Analysis

- A Future_CLV histogram showed a strongly right-skewed distribution.
- A correlation heatmap was created to examine relationships between customer features and the target.
- Boxplots and IQR-based analysis were used to examine outliers.

Target Summary Statistics

Statistic	Future_CLV
Count	2868
Mean	1038.63
Std	3215.21

Statistic	Future_CLV
Min	7.90
25%	269.08
Median	478.02
75%	945.46
Max	87261.46

7. Correlation Analysis

The strongest observed correlations with Future_CLV were Monetary (0.80), Frequency (0.60), Total_Quantity (0.55), Unique_Products (0.40), and Active_Months (0.34). Recency showed a negative relationship (-0.15). Average_Order_Value (0.04) and Avg_Purchase_Interval (-0.02) had very weak linear relationships.

Feature	Correlation with Future_CLV
Monetary	0.80
Frequency	0.60
Total_Quantity	0.55
Unique_Products	0.40
Active_Months	0.34
Recency	-0.15
Average_Order_Value	0.04
Avg_Purchase_Interval	-0.02

Interpretation: historical spending and purchasing volume were the strongest linear indicators of future customer value.

8. Outlier Analysis

Outliers were identified using boxplots and the IQR rule. They were not automatically removed because extreme observations may represent genuine high-value customers, which are important in a CLV problem.

Feature	Outlier Count	Outlier %
Monetary	307	9.27%
Frequency	243	7.33%
Recency	0	0.00%
Average_Order_Value	368	11.11%
Total_Quantity	304	9.18%
Unique_Products	242	7.30%
Active_Months	187	5.64%
Avg_Purchase_Interval	125	3.77%
Future_CLV	271	8.18%

9. Baseline Models

Simple baselines were used as benchmarks. A useful ML model should improve upon them.

Model	MAE	RMSE
Mean Baseline	955.96	3915.39
Median Baseline	820.43	3964.10

The median baseline had lower MAE, which is expected because the median is less sensitive to extreme target values.

10. Random Forest Experiments

Random Forest was used because it handles non-linear relationships and interactions well and does not require feature scaling.

Experiment	MAE	RMSE	R ²
Random Forest Baseline	644.12	2639.68	—
Random Forest + Log Features	641.85	2609.34	0.5552

The log-feature experiment produced only a small MAE improvement. Extreme customers remained difficult to model.

Random Forest importance ranked Monetary as the most important feature, followed by Total_Quantity, Frequency, and Unique_Products.

11. XGBoost Experiments

XGBoost is a gradient boosting algorithm that sequentially builds trees to correct earlier prediction errors.

Experiment	MAE	RMSE	R ²
XGBoost Baseline	663.78	2282.51	0.6596
Tuned XGBoost	616.57	2640.58	0.5445

Final model selection: Baseline XGBoost was selected because it achieved the best recorded RMSE and R². Tuned XGBoost had lower MAE but worse RMSE and R², so it was not better overall.

12. Evaluation Metrics

MAE: Average absolute difference between actual and predicted values. Lower is better.

RMSE: Square root of average squared prediction error. It penalizes large errors more strongly.

R²: Measures how much variation in the target is explained by the model. Higher is generally better.

Because Future_CLV has extreme high-value customers, RMSE was particularly important for evaluating large prediction errors.

13. Final Model Performance

Metric	Final XGBoost Baseline Result
MAE	663.78
RMSE	2282.51
R ²	0.6596

The final model demonstrated meaningful predictive capability but still showed larger errors for rare extreme-value customers.

14. Actual vs Predicted and Residual Analysis

The Actual vs Predicted plot compared true Future CLV values with predictions. Points close to the diagonal line represent accurate predictions.

The plots showed better concentration for low and medium CLV customers and larger deviations for extreme customers.

Residuals were calculated as Actual_CLV – Predicted_CLV. The residual analysis showed the largest errors around rare, high-value customers, consistent with the target's heavy right tail.

15. XGBoost Feature Importance

The XGBoost importance ranking was approximately: Monetary (0.31), Total_Quantity (0.27), Frequency (0.11), Recency (0.09), Active_Months (0.08), Unique_Products (0.06), Average_Order_Value (~0.04), and Avg_Purchase_Interval (~0.04).

Main insight: Past spending and purchase volume are the strongest predictors of future customer value.

16. Prediction Output

The final model generated a Predicted_Future_CLV value for each customer. Approximately 1,377 unique prediction values were observed.

The observed prediction range was approximately -38.55 to 61,293.36. Negative CLV predictions are not business-meaningful and should be clipped to zero in a production deployment.

Repeated predictions can occur naturally in tree-based models when customers fall into similar terminal regions of the learned trees.

17. Customer Segmentation

Segment	Meaning	Recommended Action
Low Value	Lower predicted future contribution	Use automated, low-cost engagement
Medium Value	Moderate future contribution	Cross-sell, upsell, and encourage loyalty
High Value	Highest predicted future contribution	Prioritize retention and personalized service

The displayed segmentation chart showed approximately equal-sized groups, suggesting quantile/rank-based segmentation. This creates balanced groups for analysis; business-specific absolute CLV thresholds can also be used.

18. Key Business Insights

- Monetary value is the strongest predictor of Future CLV.
- Total quantity and purchase frequency are also major drivers.
- Longer recency is negatively associated with future value.
- A small number of customers contribute disproportionately to potential future value.
- High-value customers should receive priority retention and personalized service.
- Medium-value customers provide strong opportunities for cross-selling and loyalty growth.
- Low-value customers can be managed using scalable, low-cost engagement.

19. Limitations

- Future_CLV is highly skewed and contains extreme values.
- Rare high-value customers are difficult to predict accurately.
- The model uses historical behavioural data and lacks factors such as campaigns, seasonality, and demographics.
- Predicted segments inherit model uncertainty.
- Negative regression predictions need production-level business constraints.

20. Future Improvements

- Use strictly time-based validation and time-aware cross-validation.
- Experiment with log-transforming the target.

- Compare LightGBM and CatBoost.
- Add product, seasonal, campaign, and customer lifecycle features.
- Use SHAP for detailed explainability.
- Apply business-driven CLV thresholds for segmentation.
- Clip negative predicted CLV values to zero.

21. Final Conclusion

The project successfully implemented a complete end-to-end machine learning workflow for Customer Future CLV Prediction. It covered data preparation, target creation, feature engineering, RFM analysis, EDA, outlier analysis, baseline models, Random Forest, XGBoost, evaluation, interpretation, prediction, and customer segmentation.

Baseline XGBoost was selected as the final model because it produced the strongest overall recorded RMSE (2282.51) and R^2 (0.6596). The main business finding is that historical spending, total purchasing quantity, and purchase frequency are the strongest drivers of future customer value.

Final Project Status: Completed.